Last time - we talked about how we can use data to construct estimates of quantities we're interested in.

use data to estimate quantity of interest

If we are interested in the % of overall population who are left-handed, we could estimate that using the proportion of people in a random sample who are left handed.

$\hat{p} = \frac{Y}{n}$ sample proportion estimates true p

Using a binomial probability model, we said the expected value of the sample proportion is equal to the true percentage for the overall population.

$Y \sim Binom(n, p)$ $\hat{p} = \frac{Y}{n}$ $\mathbb{E}(\hat{p}) = p$

When we go out and actually collect sample data, how close the sample proportion is to the true population percentage depends on how much data we collect. For larger samples, it's more likely that the sample proportion will be closer to the true population percentage.

$\hat{p}$ closer to p for large n

We are often interested in quantifying that closeness.

How close to the true value is the sample proportion likely to be? In other words, how precise is the sample proportion as an estimate of true p?

↳ how precise is $\hat{p}$?

To answer that, we need to think about the variance of the sample proportion.

$\hat{p} = \frac{Y}{n}$ ex: Y = # LH people out of n

We mentioned last time that the expected sample proportion equals the true population percentage:

$$\mathbb{E}(\hat{p}) = \mathbb{E}\left(\frac{Y}{n}\right) = \frac{\mathbb{E}(Y)}{n} = \frac{np}{n} = p$$

Similarly, the variance of the sample proportion is

$$\text{var}(\hat{p}) = \text{var}\left(\frac{Y}{n}\right) = \frac{\text{var}(Y)}{n^2} = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}$$

n in denom.

The variance of the sample proportion has "n" in the denominator. Larger samples → lower variance.

That agrees with what we explored in simulations previously — when the sample size is larger, the sample proportion tends to be less spread out — i.e. lower variance.

What is the actual probability distribution of the sample proportion?

distribution of $\hat{p}$?

In general, knowing the mean and variance of an RV does not tell us the actual probability of each possible outcome.

For the next part, it helps to recognize that the sample proportion is actually a type of sample average. Specifically, each person we sampled had a binary response, eg & 1 if left-handed, 0 otherwise.

$X_i = \begin{cases} 1 & \text{if LH} \\ 0 & \text{else} \end{cases}$

The total number of left-handed people in the sample (Y) equals the sum of each person's binary response.

$Y = X_1 + \ldots + X_n$

each person sampled

Then, the sample proportion is technically equal to the average of each person's binary response.

$\hat{p} = \frac{X_1 + \ldots + X_n}{n}$

The other main result we talked about last class was the central limit theorem. The central limit theorem says that for large samples, the sample average follows an approximately normal distribution

central limit theorem: $\bar{X}_n \approx$ normal for large n

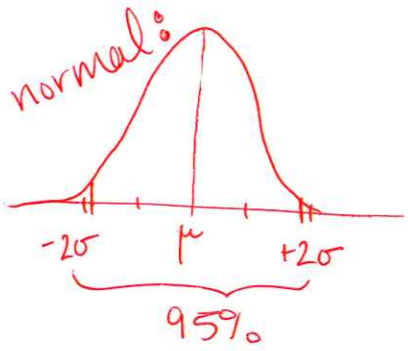
So because the sample proportion technically is a sample average, the CLT applies

$\hat{p}$ is a sample mean → CLT applies

we already worked out that its expected value is p and variance is $p(1-p)/n$. The central limit theorem tells us that it has an approximately normal dist. with that expected value and variance

$\hat{p} \overset{\text{approx}}{\sim} N\left(p, \frac{p(1-p)}{n}\right)$

For a normally distributed RV, 95% of observations fall within 2 standard deviations of the mean in either direction.



For large enough samples, the sample proportion $\hat{p}$ falls within 2 std dev. of the true population percentage approx. 95% of the time.

$$P(p - 2*\text{stdev} < \hat{p} < p + 2*\text{stdev}) \approx 95\%$$

This gives us a tool to say, if the true population percentage of people who are left-handed is 15%, then if I take random samples of 40 people, the proportion of ~~left~~ people in our sample will be between .15 − 2*stdev and 0.15 + 2*stdev 95% of the time.

What does that work out to:

$Y \sim$ Binom(40, 0.15)

population: $p = 0.15$

sample $n = 40$

$$0.15 \pm 2*\sqrt{\frac{p(1-p)}{n}}$$

$$0.15 \pm 2*\sqrt{\frac{0.15(0.85)}{40}} \approx [0.037, 0.263]$$

So ~~in~~ a sample of 40 people, the sample proportion who are left-handed should be between 3.7% and 26.3%, with 95% probability.

What if instead, we want to reason in the opposite direction: we have observed sample data, but don't know the true population proportion.

opposite direction: have $\hat{p}$ → what is true p?

We sampled 40 people, and 10 of them were left handed. What percentage of the overall population is left handed?

ex: observed $n=40$, $y=10$

$\hat{p} = \frac{10}{40}$

The sample proportion is 25%, but we know that isn't necessarily exactly equal to the true percentage.

A popular way to describe our guess about the true percentage is by giving an interval of plausible values it could be, based on the data we observed.

$p = ?$

idea: range of plausible values

And we construct an interval in an analogous way to what we did before:

$\hat{p}$ within $\pm$ 2 stder of p 95% of time

We know that $\hat{p}$ is within 2 stder of true p with 95% probability, so it's reasonable to guess that the true p is within +/- 2 stder of the sample proportion we observed.

guess: p within $\pm$ 2 stder of $\hat{p}$

One issue with this is that the standard deviation depends on the true population percentage p, which we don't know

issue: $\text{var}(\hat{p}) = \frac{p(1-p)}{n}$ → don't know p

Instead, we can approximate it by plugging in the sample proportion.

approx: $\text{var}(\hat{p}) \approx \hat{p}(1-\hat{p})/n$

$\hat{p} \pm 2^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ ← which gives us this interval for our guess for the true value of p.

For our sample of 40 people, of whom 10 were left handed, what does this work out to?

$0.25 \pm 2^* \sqrt{\frac{0.25(0.75)}{40}}$

$\approx [0.11, 0.39]$ This is a "confidence interval", and we might called "(95%) confidence interval"

say we are 95% "confident" that between 11% and 39% of the overall population are left-handed.

The mathematical meaning of this can be a little confusing.

"95% confident" ≠ "95% probability"

% LH people not random

↳ either in [.11%, 39%.] or not

The true percentage of LH people is some fixed number. We don't know it, but it's a fixed value. That fixed number is either between 11 and 39%, or it isn't. We don't say there is a 95% prob. that the true percentage is between 11 and 39%.

recall: probability as long-run frequency

We said that probability corresponds to the long-run frequency of an outcome over many repetitions. There is no corresponding "long-run frequency" of the true %.

of left handed americans being between 11% and 39%.

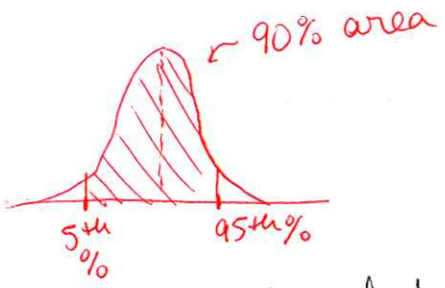
BEFORE collecting data:
95% prob. that we get data values that result in CI that includes true p

BEFORE we go out and collect any data, from a theoretical standpoint, there is a 95% chance that the data values we observe will result in a confidence interval that includes the true value.

The ±2 stdev corresponds to a 95% confidence interval because the normal distribution has 95% of its probability within 2 stdev of its mean.

95% CI : ±2 σ

other levels?



If we want a 90% confidence interval, we need to check how many standard deviations away from the mean we need to go in order to capture 90% of the probability for a normal distribution.

↳ If the central chunk is 90%, its endpoints are the 5th and 95th percentiles. What is the 95th percentile?

qnorm(0.95, mean = 0, sd = 1) ≈ 1.645

To get a 90% CI, we need to go 1.645 stdev away from the mean in each direction

$$\hat{p} \pm 1.645 * \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

For example, a 90% CI for the true population % of left-handed people would be (based on y=10 & n=40)

$$0.25 \pm 1.645\sqrt{\frac{0.25(0.75)}{40}} \approx [0.14, 0.36]$$

We are 90% confident that the overall percentage of people who are left handed is between 14% and 36%.

CI for means

Instead of proportions, we can also construct confidence intervals for the mean of some continuous-valued quantity.

X = BP reduction due to drug

$X \sim p(x)$ (arbitrary)

$E(X) = \mu$ unknown $\text{var}(X) = 4$

Taking a medication for high blood pressure causes a reduction in BP, but the actual amount of the reduction varies from person to person.

Suppose that across the entire population, the BP reduction follows some dist. with unknown mean μ and known stder = 2. (For now we assume we know the standard deviation, which is not very realistic, but we'll address that later.)

We randomly select 50 people, they take the medication, and we measure the reduction in their blood pressure.

observe $X_1, X_2, \ldots, X_{50}$

$\bar{X}_{50} = 8$

Across those 50 people, the average reduction in systolic BP was 8 mmHg.

What can we say about μ, the population average effect of the medication on BP?

We start by considering the sampling distribution of the sample average $\bar{X}_{50}$

dist of $\bar{X}_{50}$

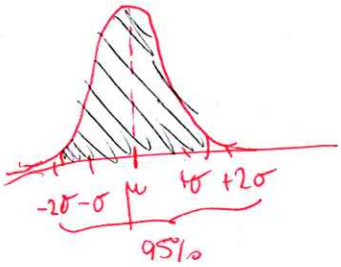
The central limit thm tells us that $\bar{X}_{50}$ is approximately normally distributed...

$\bar{X}_{50} \overset{\text{approx}}{\sim} N\left(\mu, 2^2/50\right)$ ... with expected value μ

central limit thm

and variance equal to the original variance divided by 50, the sample size

Then, just like we said earlier when talking about proportions, we know that a normally distributed RV takes values within +/- 2 stdev of its mean with 95% probability.



The sample avg of 50 people is within 2stdev of the true population mean, μ, 95% of the time.

in our example: $2 * \sqrt{2^2/50} \approx 0.57$

$\bar{X}_n \in [\mu \pm 2\sigma$

$\bar{X}_n \in [\mu \pm 0.57]$ w/ 95% prob

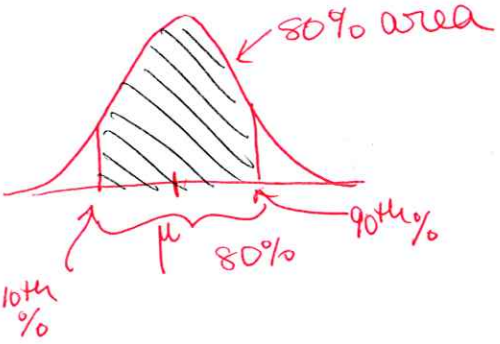
Based on that, a plausible range for the true population mean μ, is

$\bar{X}_{obs50} \pm 0.57$ 95% CI for μ

Based on the observed data, a 95% confidence interval for μ is

$\approx [7.4, 8.7]$

Just like before, if we want a different level of confidence interval, we use the quantiles of the normal distribution



If we want an 80% confidence interval, how many standard deviations away from the mean do we need to go?

↳ central chunk is 80%

↳ its endpoints are the 10th and 90th percentiles

what is the 90th percentile?

qnorm(0.9, mean=0, sd=1) ≈ 1.28

To capture 80% of the probability of the normal dist, we need to go 1.28 standard deviations out from the mean in each direction.

80% CI for μ : $8 \pm 1.28 * \sqrt{\frac{2^2}{50}}$

So our 80% CI for the true avg population effect of the BP medication is

≈ [7.6, 8.4]

We started this example by assuming we knew the true variance of the population distribution.

assumed known var(X) for population dist

↳ usually DON'T know this

In practice, we generally won't know that value. What do we do?

estimate var(X) from data, using the sample variance,

$S_x^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{X}_n)^2$ sample variance ← calculated as

sample stdev = $\sqrt{S_x^2}$

Having to also estimate the variance from the data adds extra uncertainty.

also guessing variance : adds extra uncertainty

Because of that extra uncertainty, we need to make our confidence intervals wider to still have a 95% chance of covering the true value of μ.

extra uncertainty → wider interval to keep 95% chance of covering μ

How much wider?

To get the right width, instead of using quantiles of the normal dist, we use quantiles of a different dist., called the "t" distribution

known var(x)
percentiles of normal dist → unknown var(x) percentiles of "t" dist

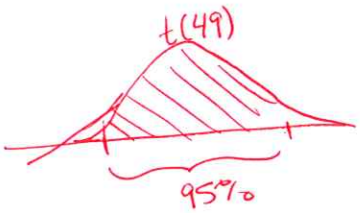
The t distribution is another FAMILY of distributions.

The parameter you need to specify for the t distribution is called "degrees of freedom"

t dist
* family
* parameter: "degrees of freedom" = n-1 in our example

The "degrees of freedom" to use when constructing a confidence interval for a mean, is equal to n-1, where n is the # of data points.

n = 50 → df = 49



For a t dist with 49 degrees of freedom, how many stder away from the mean do we have to go in order to capture 95% of the probability? Check the 97.5th percentile of the t dist:

qt(0.975, df=49) ≈ 2.01

observation: that was almost no different than the normal distribution quantile

*(technically it's ≈1.96 for Normal, not exactly 2)

* t distribution looks very similar to Normal when "degrees of freedom" value is large.
  - ↳ df large when n is large – lots of sample data
  - ↳ when we have a large amt of data, we can do a pretty good job estimating the variance from the data, so it doesn't add much extra uncertainty